Higher National Diploma in Civil Engineering / Quantity Surveying

First Year, Second Semester Examination – 2018

CE1205 / QS12044- Civil Engineering Construction I

Instructions for Candidates:
Answer any **Five (05)** questions.
All question carry equal marks

No. of questions : 06
No. of pages : 02
Time : Three (03) hours

1.

a. State any four basic areas of Civil Engineering. (4Marks)
b. State any four basic requirements of a building. (4Marks)
c. State any four classifications of buildings with example (6Marks)
d. Briefly explain the various types of excavations (6Marks)

(Total 20 Marks)

2.

a. Write down any four basic components of a building? (4Marks)
b. Discuss the main requirement of each part stated in (b.) to fulfill its primary function. (6Marks)
c. Discuss the primary types of design loads acting on the foundation. (6Marks)
d. Draw a neat sketch diagram for random rubble wall foundation (4Marks)

(Total 20 Marks)

3.

a. State the types of pad footing and briefly explain one of them. (5Marks)
b. Write short notes on "hard core filling" to the plinth level (5Marks)
c. If you are instructed to supervise the brick masonry construction at a site, write the points to be considered in the above construction (5Marks)
d. Briefly explain the following terms in brick masonry.
   i. Standard size of brick (BS) and indicate "stretcher" and "header" (3Marks)
   ii. King closer of a brick (2Marks)

(Total 20 Marks)

4.

a. Describe causes of dampness and the effects of dampness in a building. (6Marks)

b. What is meant by 'Sewage" (3Marks)

c. Briefly describe types of sewer system based on ownership (4Marks)

d. Define the scaffolding. (3Marks)

e. Briefly explain the main forms of scaffolding (4Marks)

(Total 20 Marks)

5.

a. Differentiate the solid ground floor and suspended ground floor (5Marks)

b. Write down types of floor finishes with its usage. (6Marks)

c. Compare any two types of roofs including advantages and disadvantages. (5Marks)

d. State types of doors based on its method of opening. (4Marks)

(Total 20 Marks)

6.

a. Briefly explain the role of following personnel in a construction.

i. Consultant (2Marks)

ii. Client (2Marks)

iii. Site Engineer (2Marks)

iv. Quantity Surveyor (2Marks)

b. State the uses of following plants in a construction

i. Poker vibrator (2Marks)

ii. Excavator (2Marks)

iii. Scraper (2Marks)

iv. Concrete pump car (2Marks)

c. State four safety precautions taken in to account during construction. (4Marks)

(Total 20 Marks)

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